

Comparison with Flike

A comparison was made with *Flike* (Kuczera G. , 1999), which is a Bayesian flood frequency analysis software developed by Professor George Kuczera from the School of Civil Engineering at the University of Newcastle, Australia. *Flike* is compliant with the recent major revision of Australian industry guidelines for flood estimation, documented in the update of Australian Rainfall and Runoff (ARR). *Flike* uses a novel importance sampling approach for estimating the posterior rather than Bayesian MCMC. In addition, *Flike* samples prior distributions using a multivariate Normal distribution, with the default priors set to have very large variances in order to make them uninformative.

There are a number of self-training examples on the *Flike* website⁷. The examples most comparable with RMC-BestFit are examples 3 through 6. Example #3 demonstrates a flood frequency analysis using the procedures described in Australian Rainfall and Runoff Book 3: Peak Discharge Estimation⁸. Specifically, this example covers the fitting of a LPIII distribution to an annual maximum series for the Hunter River at Singleton. Results are shown in Figure 55. *Flike* does not output the posterior mode; however, it does provide the posterior predictive distribution and the posterior mean quantile curve. For this example, a comparison was made with the credible intervals and the posterior predictive distribution. As can be seen, RMC-BestFit and *Flike* produce the same results.

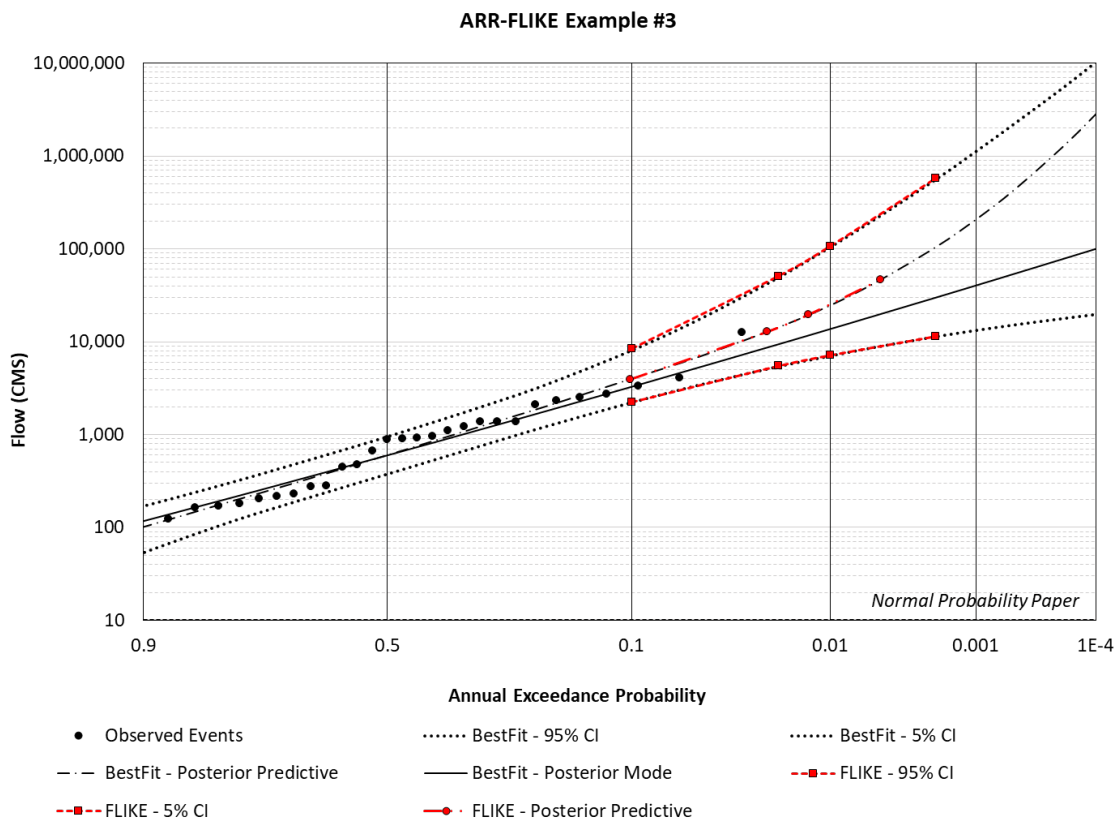


Figure 55 – Comparison of RMC-BestFit and Flike for Example #3.

Example #4 is a continuation of Example #3 and it examines the benefit of using binomial censored historical flood information. RMC-BestFit doesn't permit right censored thresholds. Nevertheless, RMC-BestFit can still replicate the binomial-censored example using an interval and setting the upper bound to be extremely large, approaching infinity. Results for Example #4 are shown in Figure 56. RMC-BestFit and *Flike* produce the same results for this example.

⁷ <https://flike.tuflow.com>

⁸ <http://arr.ga.gov.au/arr-guideline>

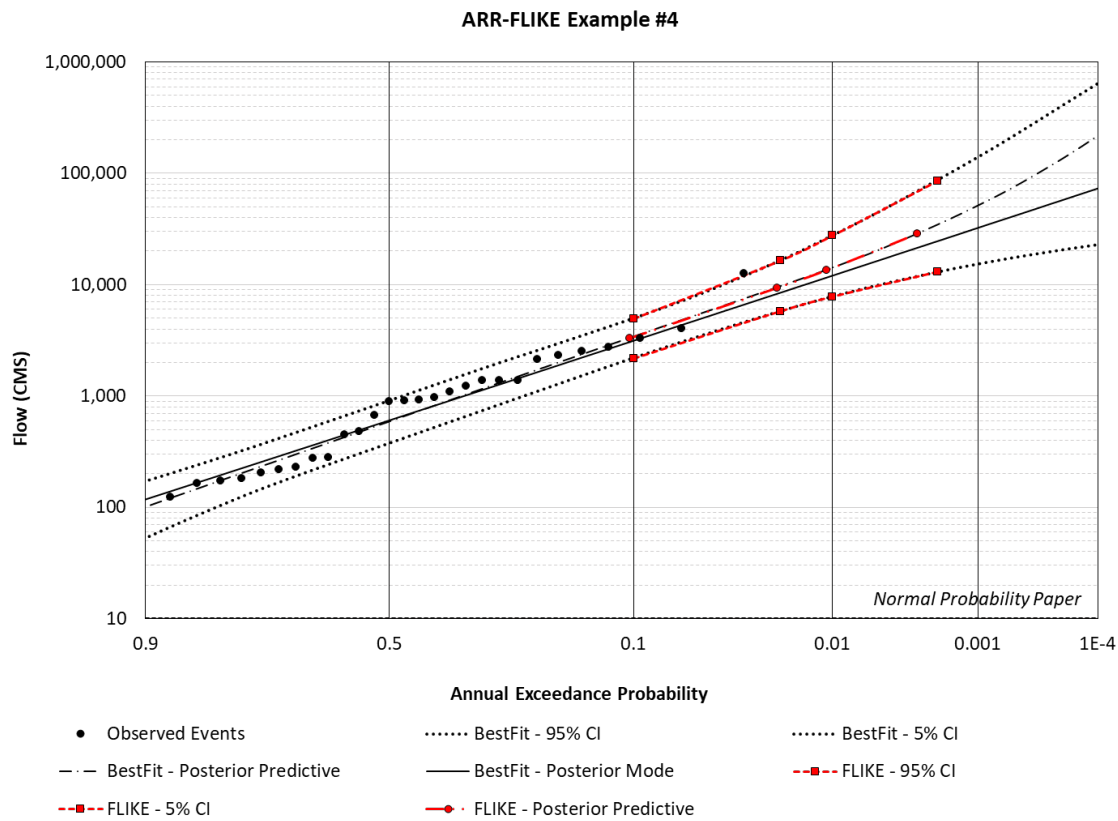


Figure 56 – Comparison of RMC-BestFit and Flike for Example #4.

Example #5 examines the use of regional information, building on Example #3. In this example, a regional skew analysis was performed and the regional skew was estimated to be 0.00 with a MSE of 0.09. This information was incorporated into the Bayesian analysis by setting the prior for the skew parameter of LPIII to be Normally distributed with a mean of 0.00 and standard deviation of 0.30. The *Flike* website only provides results for the credible intervals and posterior mean quantile for this example. Therefore, only the credible intervals were compared. As shown in Figure 57, RMC-BestFit and *Flike* produce the same credible intervals for this example.

Example #6 is a two-part example demonstrating censoring using the MGBT. The example uses 56 years of annual maximum discharges for the Wimmera River at Glynwylin. In the first part of this example, the GEV distribution was fit to the at-site data without removal of low outliers. The *Flike* website only provides results for the credible intervals for this example. Results are shown in Figure 58 below. RMC-BestFit and *Flike* results are equivalent. Next, the MGBT test was used to remove low outliers and replace them with a left-censored threshold. This approach is consistent with Bulletin 17C (U.S. Geological Survey, 2018), and is implemented in *HEC-SSP* and RMC-BestFit. The results for part two of Example #6 are shown in Figure 59. Again, RMC-BestFit and *Flike* produce the same results.

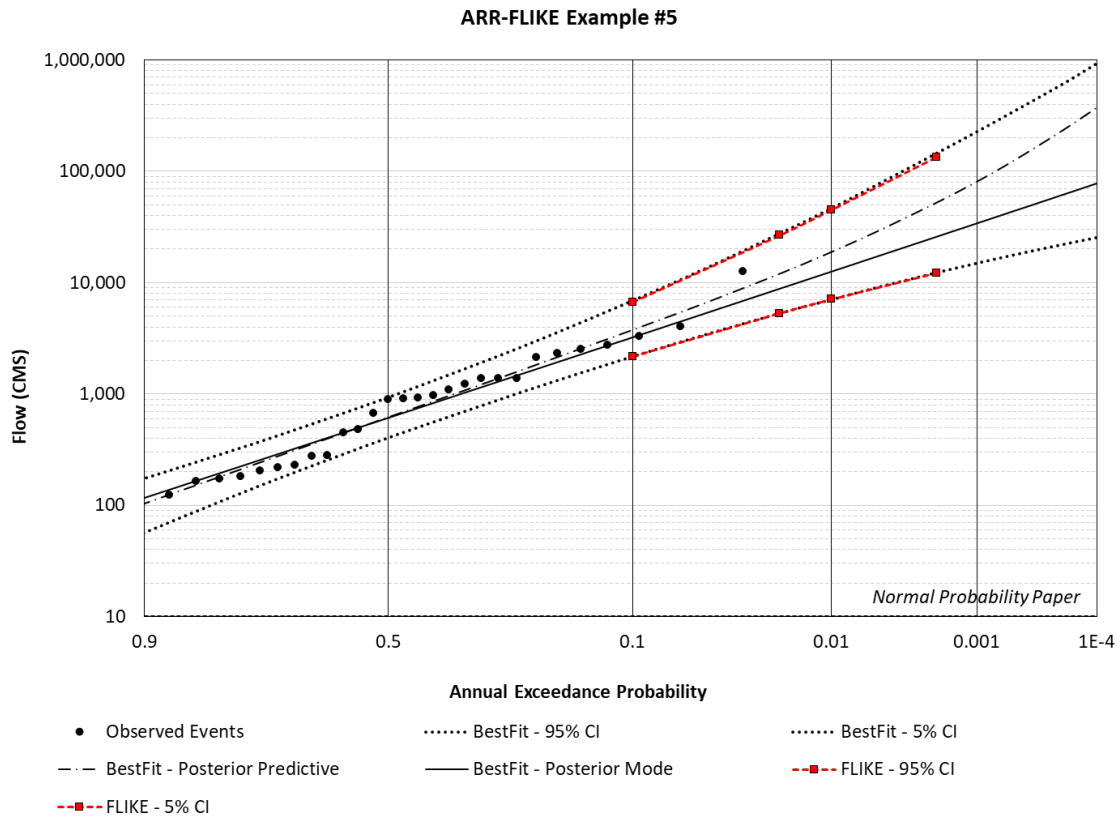


Figure 57 – Comparison of RMC-BestFit and Flike for Example #5.

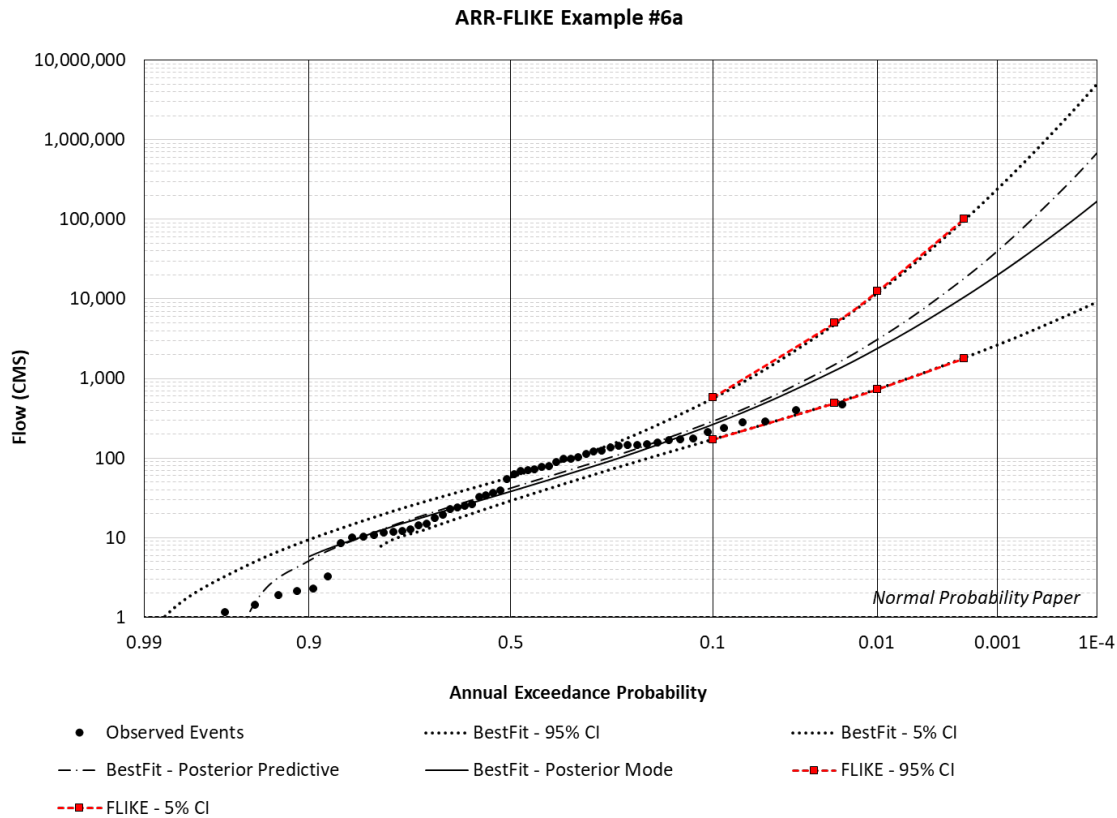


Figure 58 – Comparison of RMC-BestFit and Flike for Example #6a.

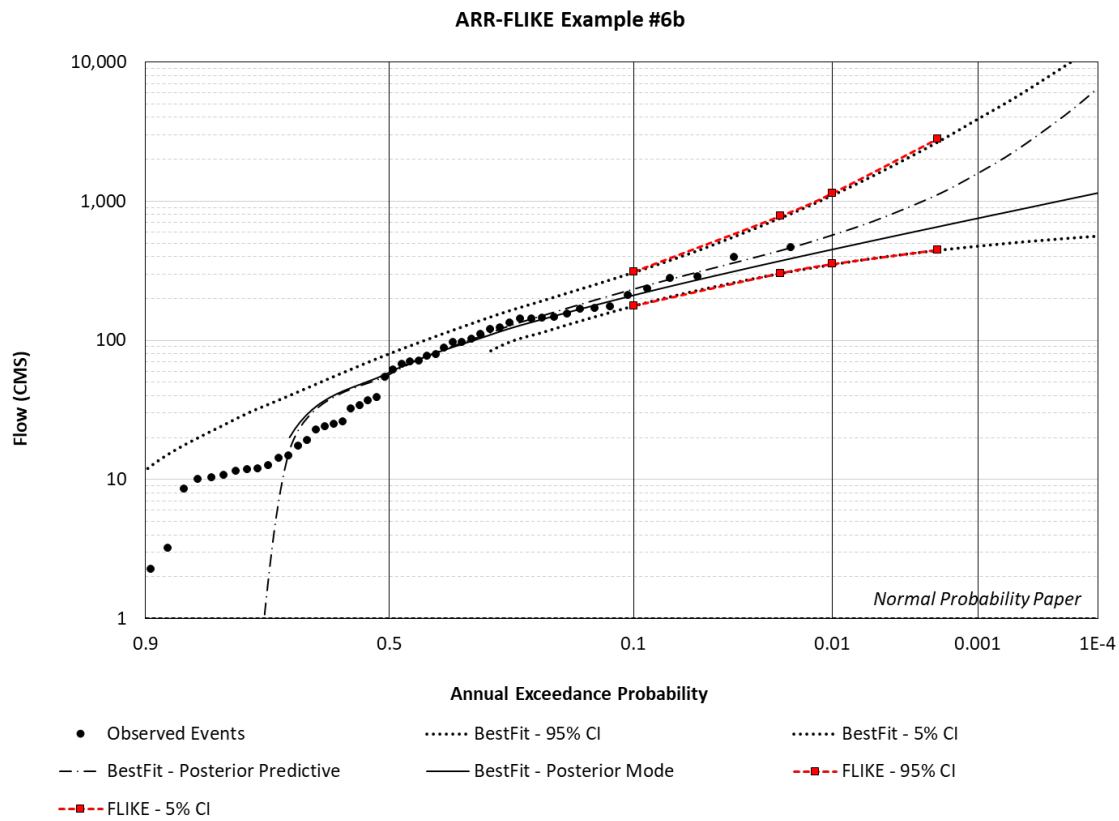


Figure 59 – Comparison of RMC-BestFit and Flike for Example #6b.